



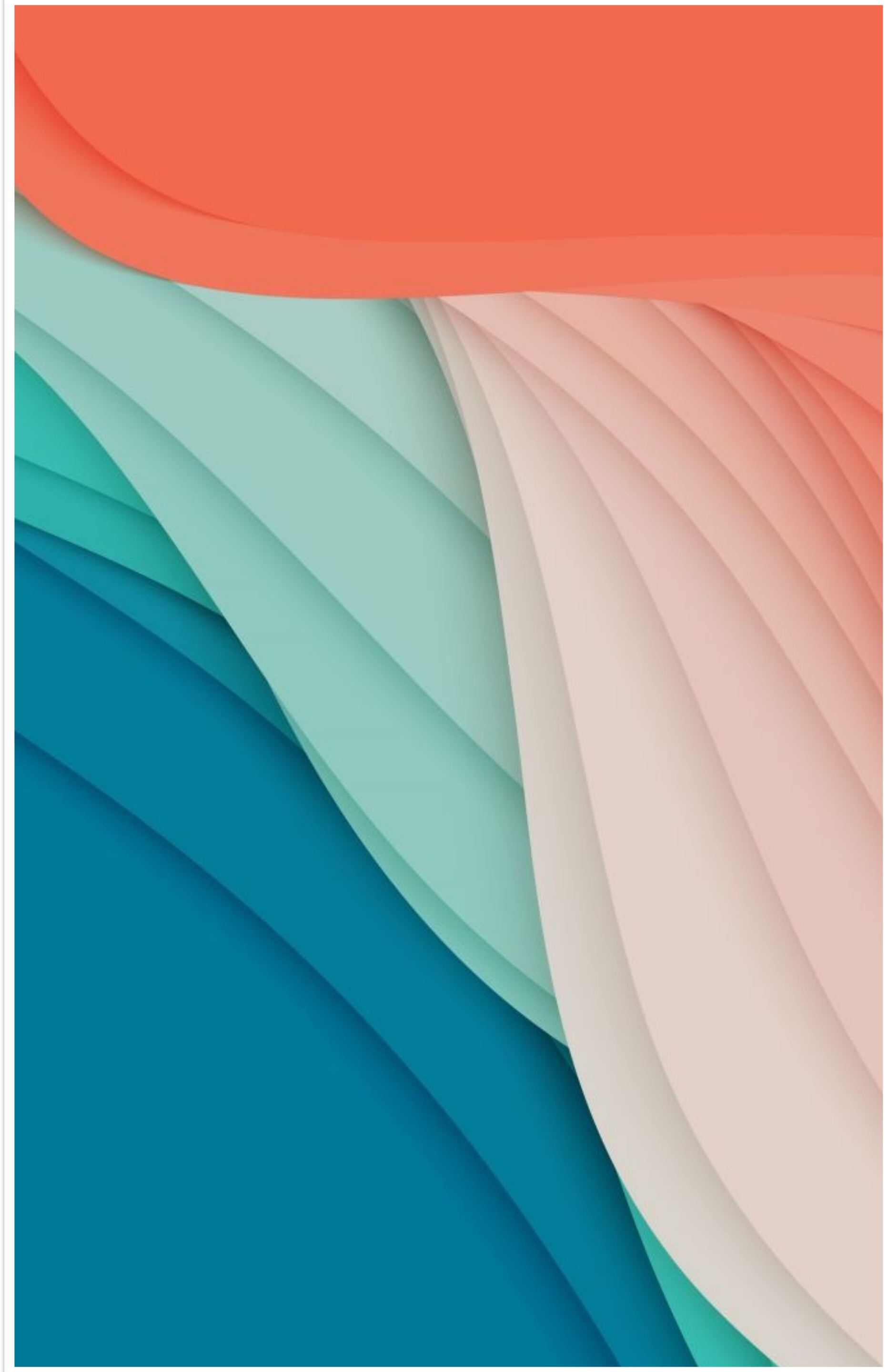
## Git - Part I

# Introduction & Basics

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# Contenu du cours

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## 1. Git et github

- a. Initiation à git : commandes de base.
- b. Utilisation de Visual Studio Code.
- c. Branches dans git.
- d. Collaboration dans git (GitHub).

## 2. Docker

- a. Bases de docker.
- b. Utilisation de docker hub.
- c. Automatisation avec docker-compose.



**Projet**



**Projet final**



**Projet final 2**



**Projet final 3**



**Projet final final**



**Projet final finaal**



**finaaaaaaaaaaal**

# What is Git ?

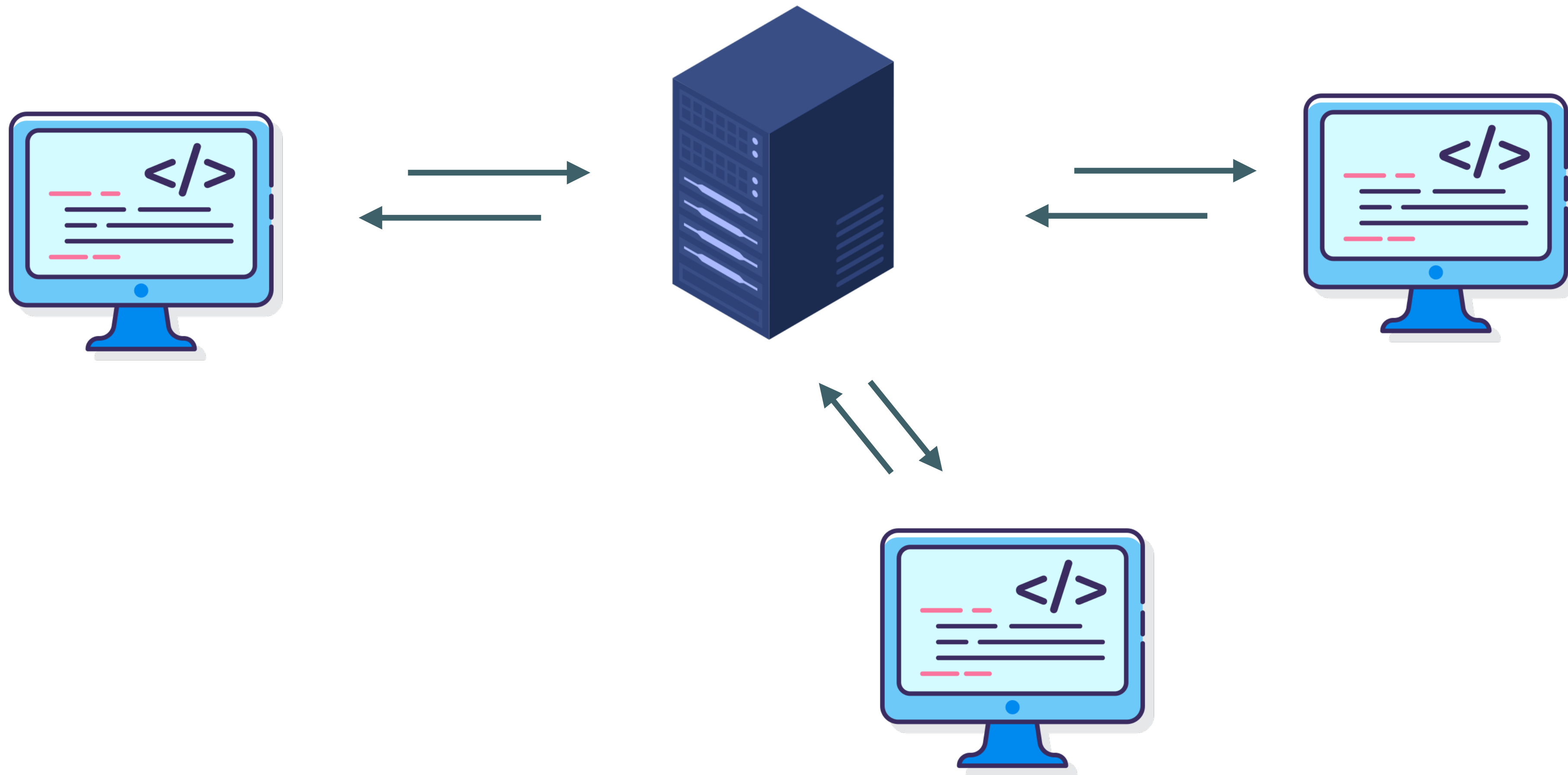
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- ◆ Git is a **distributed** version control system (VCS).
- ◆ Track changes made on files and revert back to them (like a time traveling machine).
- ◆ Developed and released in 2005 by Linus Torvalds, the creator of Linux.
- ◆ It is a free and open source.
- ◆ According to a [Stack Overflow developer survey](#) over 87% of developers use Git.

# What is a distributed VCS

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*No single point of failure*



# Git hosting service

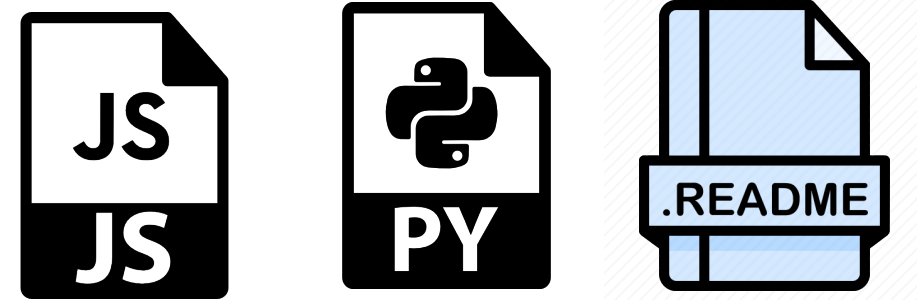
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- ◆ Web hosting service that hosts git projects on the cloud.
- ◆ The most used ones are : Github, bitbucket and gitlab.
- ◆ **Github** is the most popular one, it was bought by Microsoft in 2018.

# General git workflow

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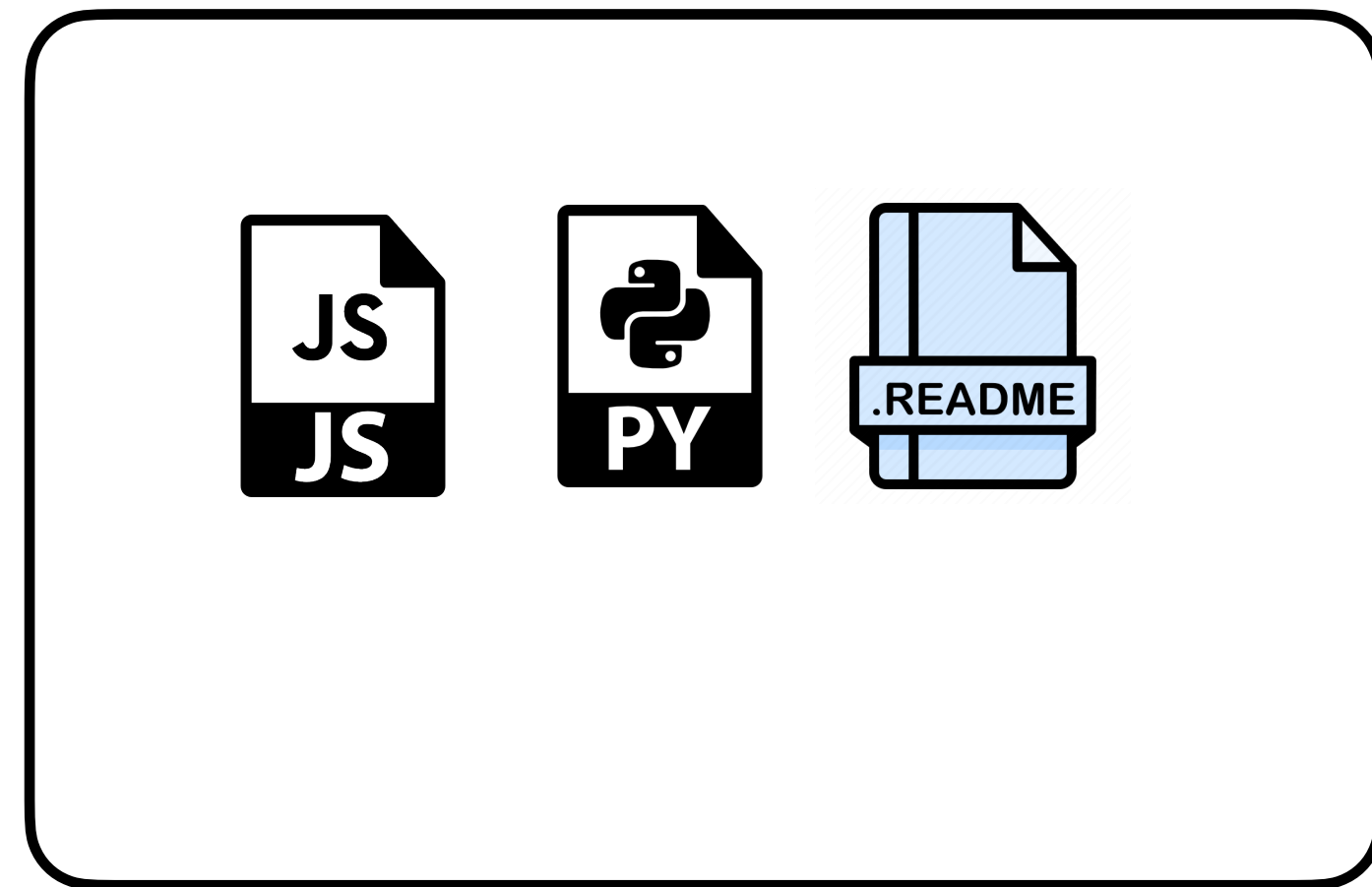
## Working directory



# General git workflow

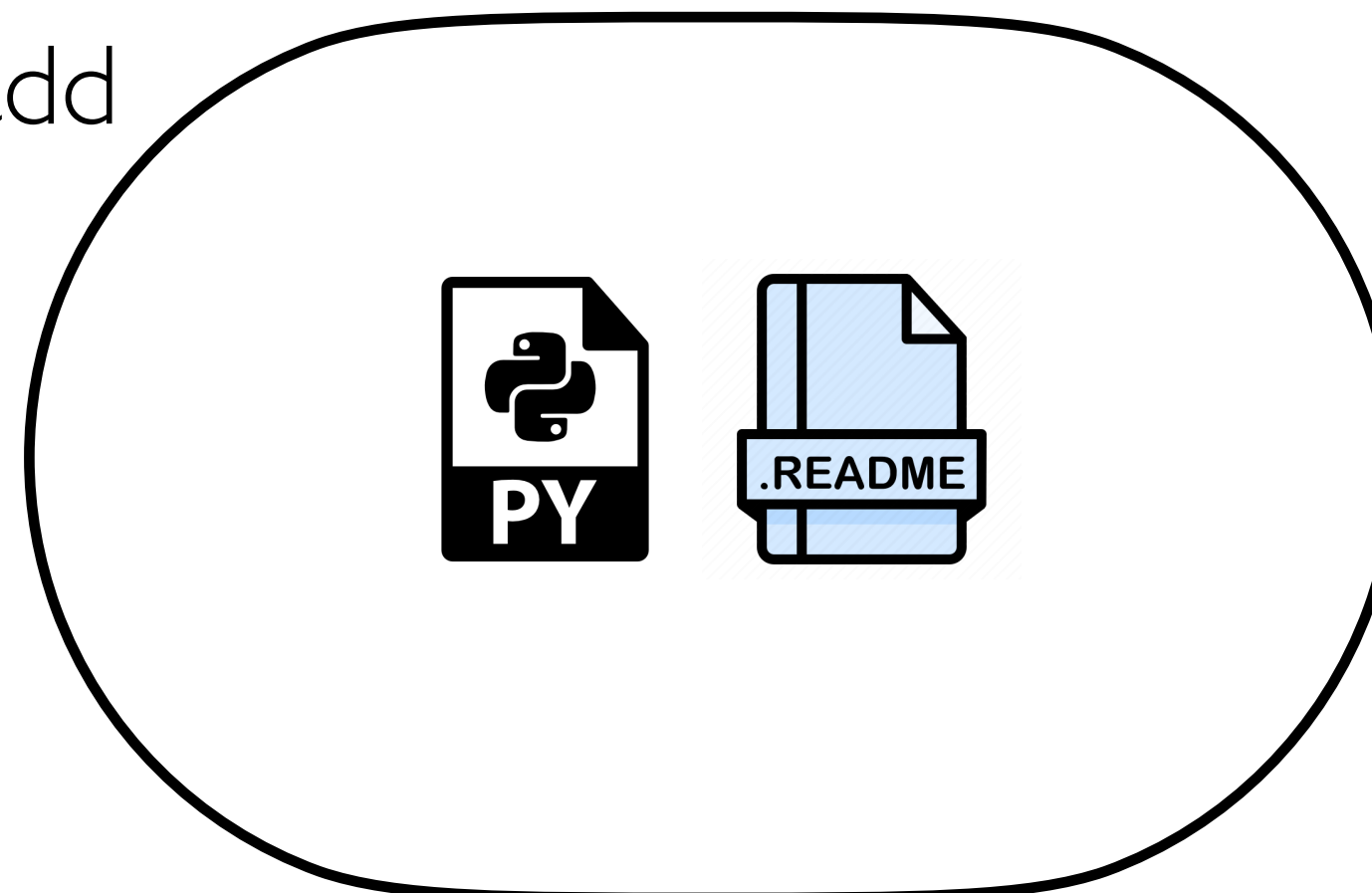
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## Working directory



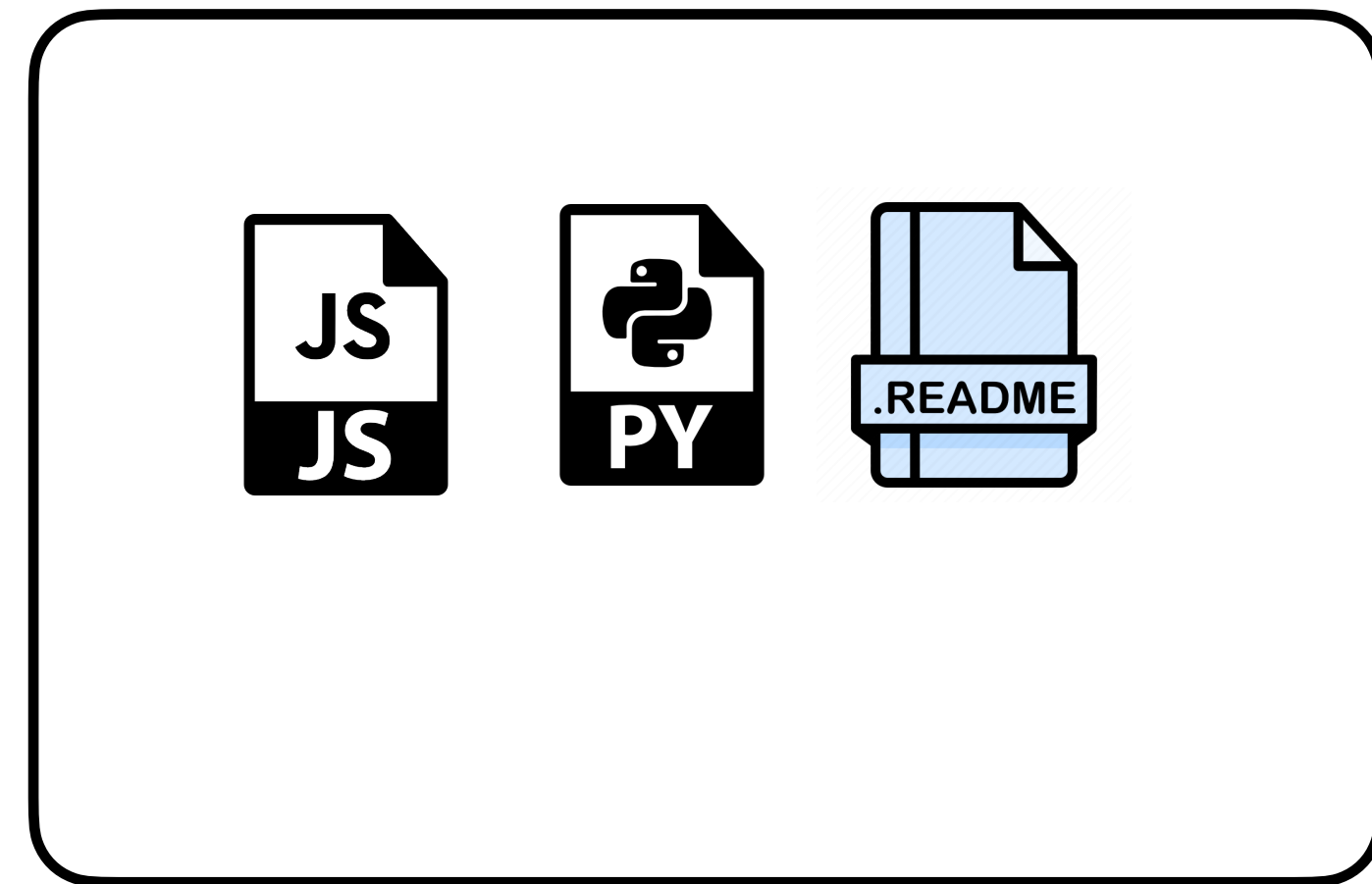
Git add

## Staging or Index (old name)



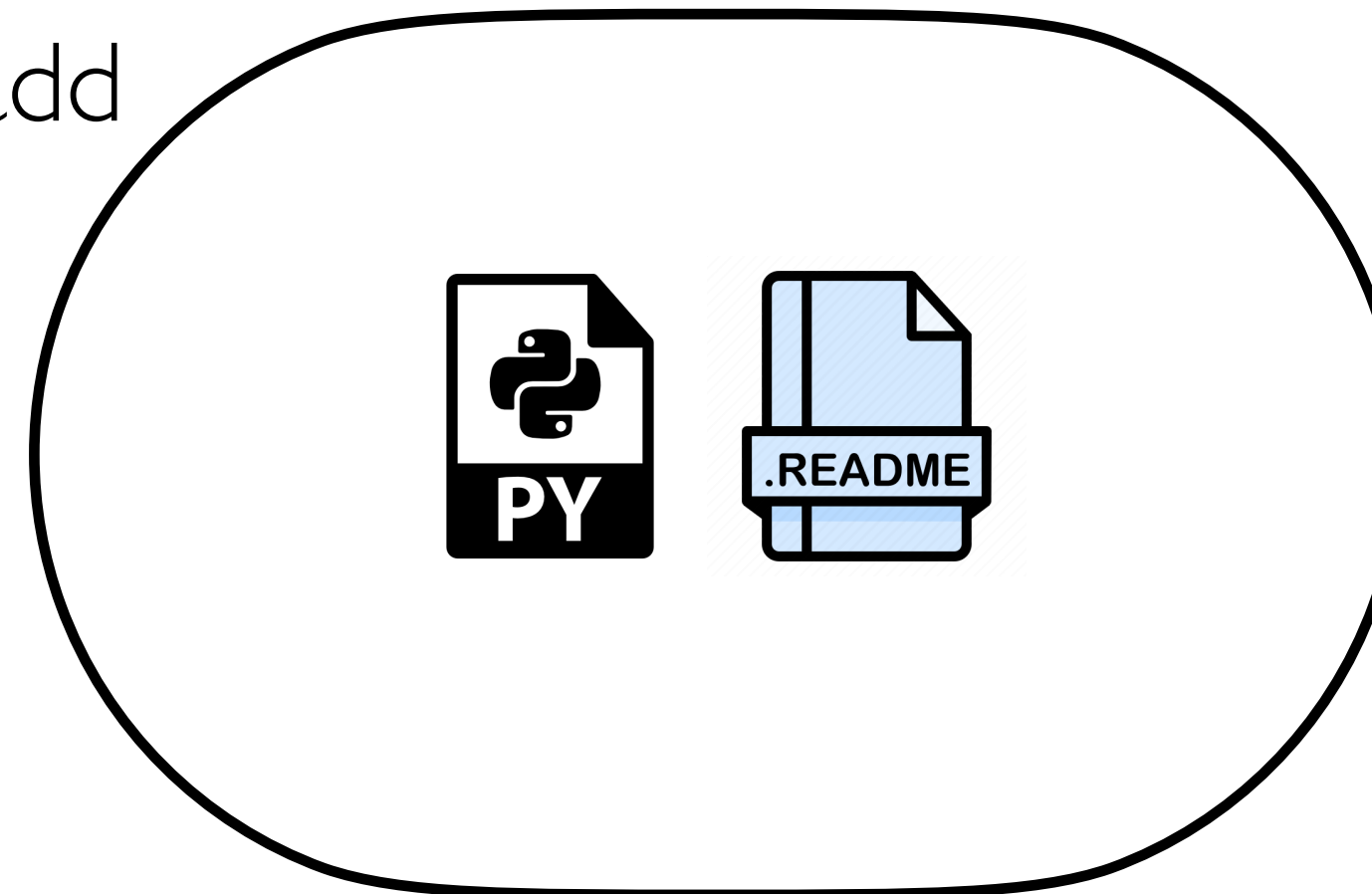
# General git workflow

## Working directory

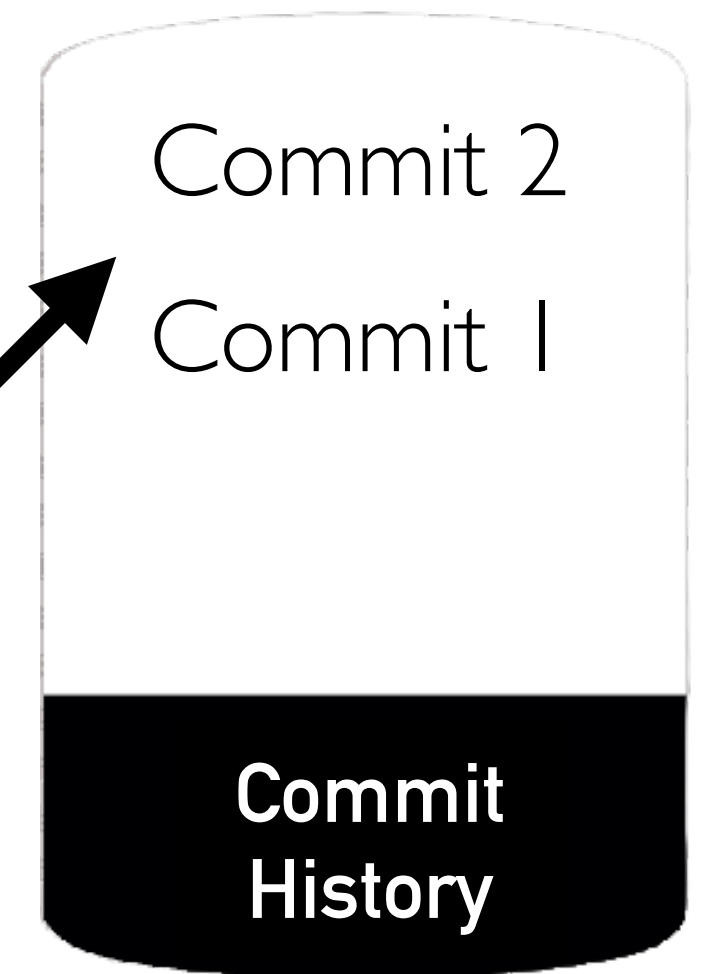


Git add

## Staging or Index (old name)



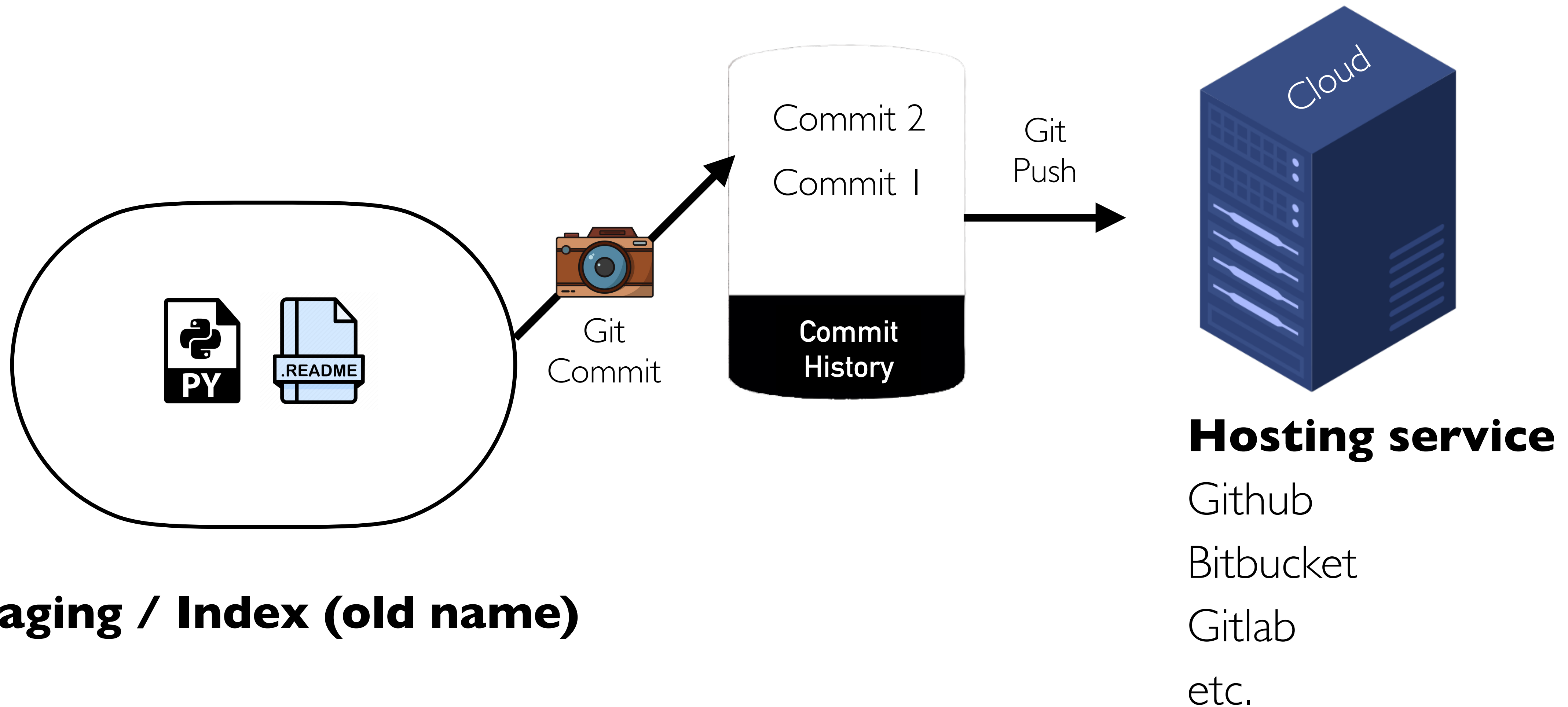
Git  
Commit



Ps. Git doesn't save the same files for each commit. Files are compressed and saved uniquely

# General git workflow

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# Git installation

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- ◆ Installation instructions : <https://github.com/git-guides/install-git>
- ◆ For windows use Git BASH

```
$ git version  
git version ****
```

# Git configuration

Cf. <https://git-scm.com/book/sv/v2/Customizing-Git-Git-Configuration>

```
$ git config [config level] [config name] [config value]
```

## ◆ [config levels] :

--local : (default) applies to the local repository, config is found in .git/config

--global : applies to the OS user repositories, config is found in ~/.gitconfig

--system : applies to the all OS users, config is found in /etc/gitconfig

## ◆ [config Name] :

user : email | name

core : editor

alias : git command

```
$ git config --global user.name 'amine'
```

```
$ git config --global user.email 'youremail@domain.com'
```

```
$ git config --local alias.st status -s
```

```
$ git config --global alias.lg 'log --oneline --graph --all'
```

```
$ git config --global -e #To see the current configuration
```

# Bash commandes reminder

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\$ cd

\$ ls -A

\$ pwd

\$ chmod

\$ grep

\$ mv

\$ cp

\$ rm -r

\$ mkdir

\$ touch

\$ cat

\$ diff

\$ echo

\$ man

\$ history

\$ clear

\$ locate

\$ open .

\$ exit

\$ kill

\$ head

\$ tail

\$ ps